Nagar YuwakShikshanSanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

Nagar YuwakShikshanSanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session	**2026-27 (ODD)**	**Course Name**	**PE-V : Lab : Blockchain Technology**
Semester	**7**	**Course Code**	**23IOT1748**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

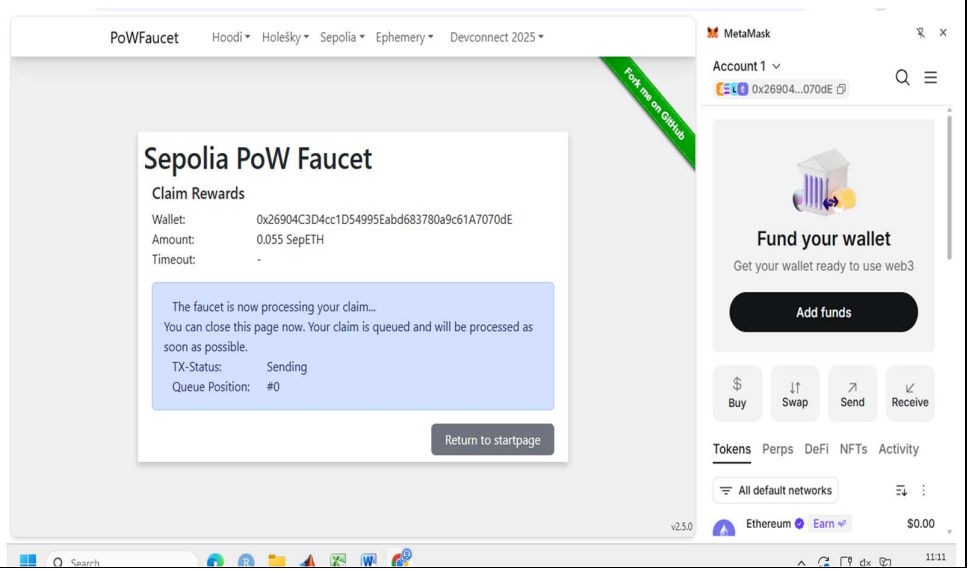
Practical Number	**09**
Course Outcome	**1. Demonstrate basics of distributed systems and cryptographic primitives used in blockchain.** **2. Outline various research aspects for development of blockchain.** **3. Make use of different security protocols to build secure blockchain applications.** **4. Analyze smart contracts and Hyperledger Fabric used in blockchain. 5. Determine the performance of applications developed with Bitcoin and Ethereum blockchain.**
Aim	Develop and Deploy a Simple Voting Smart Contract Using Solidity and MetaMask.
Theory (100 words)	A Voting Smart Contract is a blockchain-based program designed to perform the voting process in a secure and transparent manner. The contract is developed using **Solidity**, a programming language used for creating smart contracts on the Ethereum blockchain. It contains functions for selecting candidates, recording votes, and checking the total votes received by each candidate. The voting information is stored on the blockchain, making the process transparent and resistant to unauthorized changes. **Remix IDE** is used to write, compile, and deploy the Solidity smart contract. After compilation, the contract can be deployed on an Ethereum-compatible network. For this practical, **MetaMask** is used as a blockchain wallet to connect the user with the network. It manages the user's blockchain account and is used to approve and send transactions generated when a vote is submitted. When a voter selects a candidate, a transaction is created and sent to the smart contract through MetaMask. The contract verifies the transaction and updates the vote count for the selected candidate. The updated results can then be viewed through the contract functions. This practical helps in understanding the basic working of **Solidity, Ethereum transactions, Remix IDE, MetaMask, and decentralized applications (DApps)**.

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

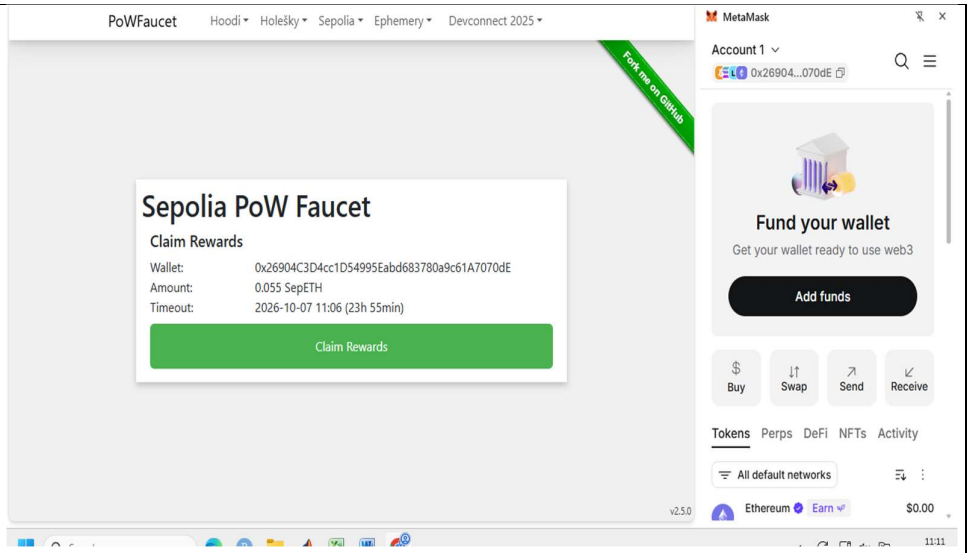
Procedure and Execution (100 Words)	1. Open Remix IDE in a web browser and create a new Solidity file named Voting.sol. 2. Write the Solidity program for the voting system with two candidates and functions for casting and checking votes. 3. Select the required Solidity compiler version and click Compile Voting.sol to check the program for errors. 4. Install the MetaMask browser extension and create or connect a wallet. Select the required Ethereum-compatible network. 5. In Remix IDE, open the Deploy & Run Transactions section and select Injected Provider – MetaMask as the environment. 6. Connect Remix with MetaMask and confirm the connection request in the MetaMask wallet. 7. Click Deploy in Remix. Confirm the deployment transaction in MetaMask and wait for the transaction to be completed. 8. After successful deployment, the voting contract appears under Deployed Contracts in Remix. 9. Select the candidate number or name and use the vote() function to cast a vote. Confirm the transaction through MetaMask. 10. Use the votes1 and votes2 functions to check the number of votes received by each candidate. 11. The candidate with the higher number of votes is considered the winner. The voting process and results are recorded on the blockchain.
	Stepwise Screenshots: 

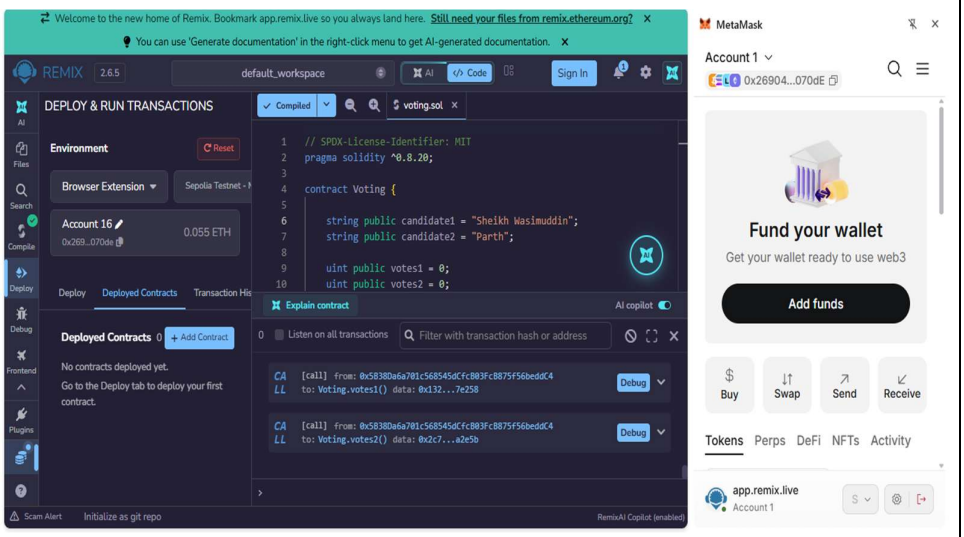
Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.





Nagar YuwakShikshanSanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

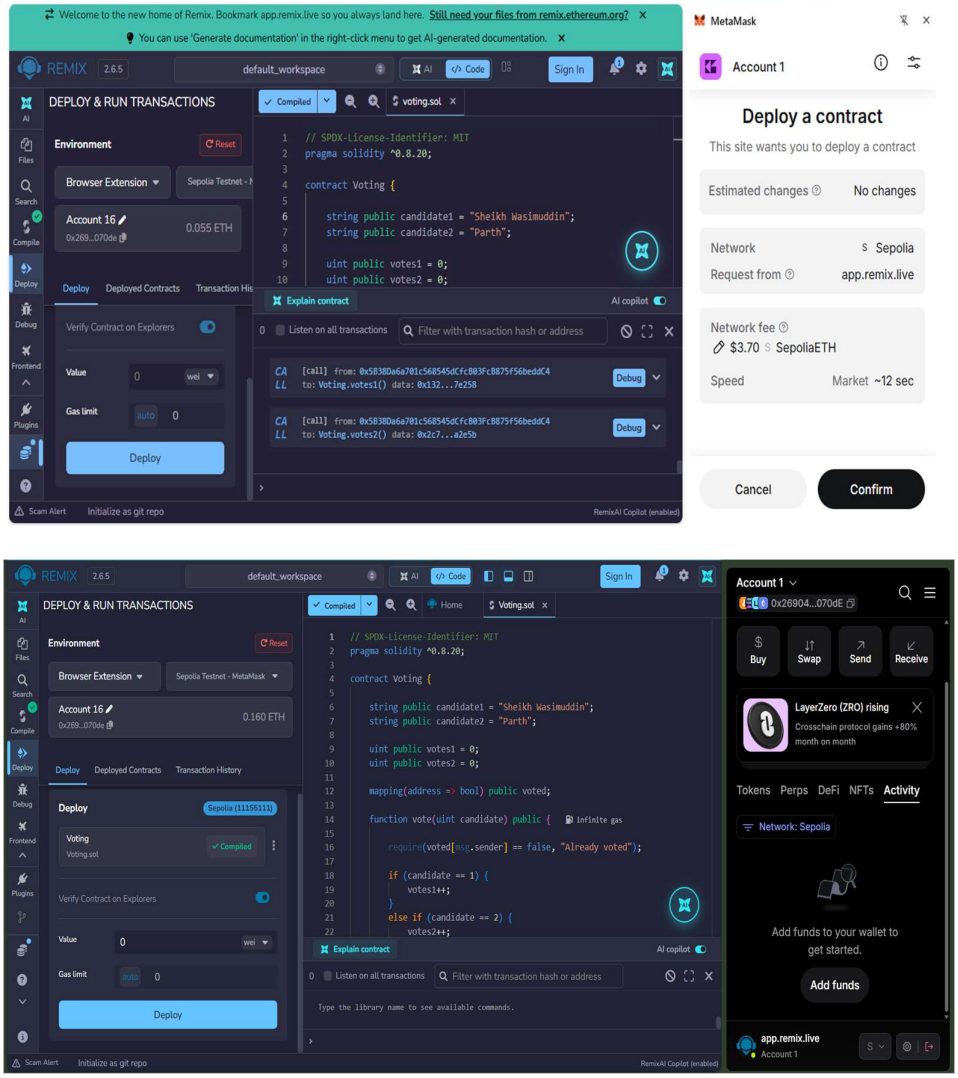
Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

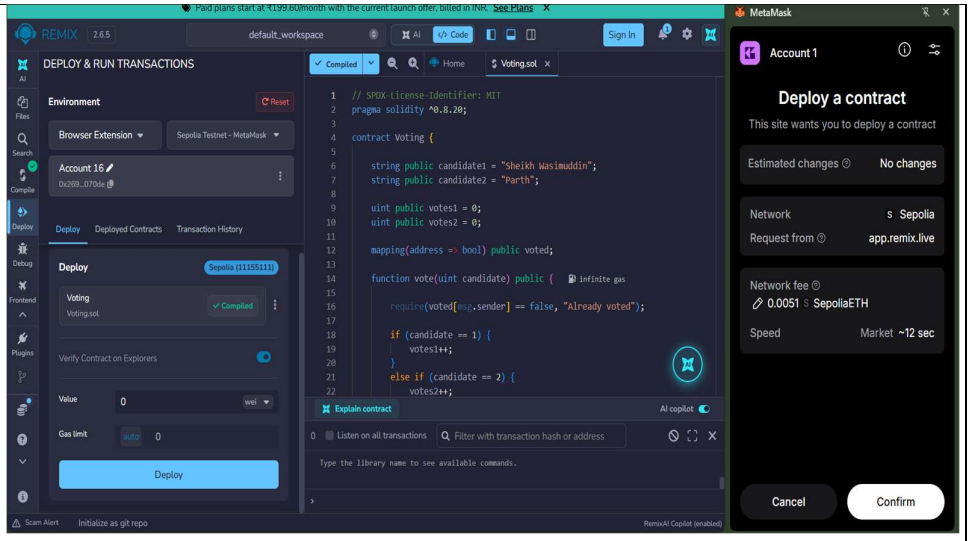


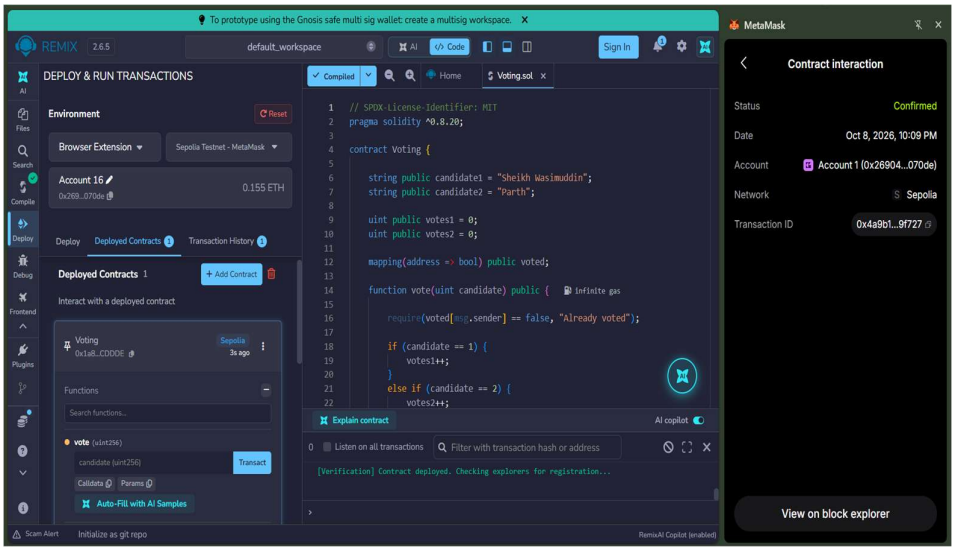
Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.



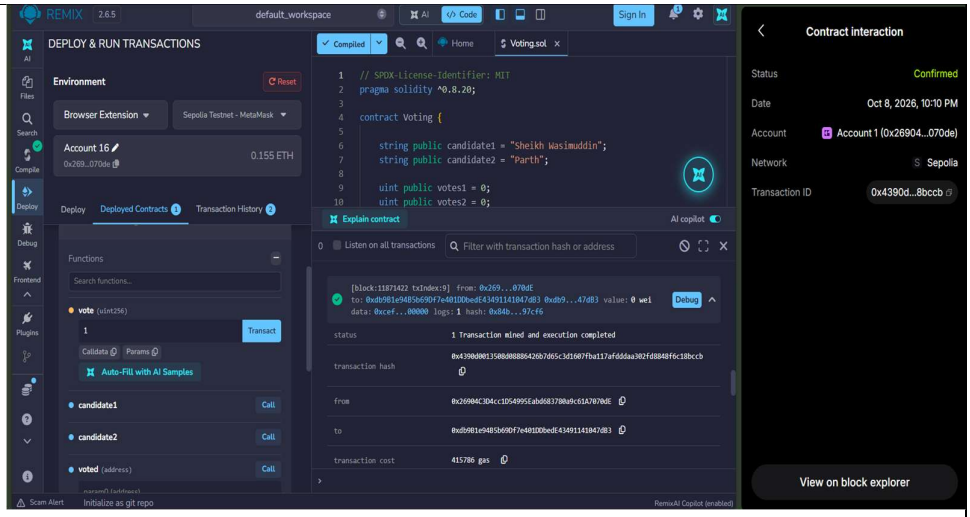


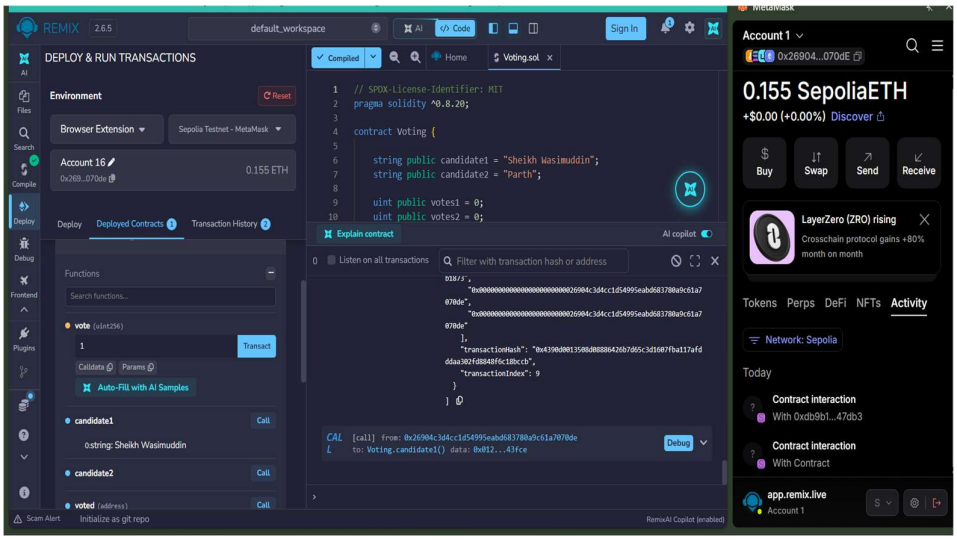
Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.



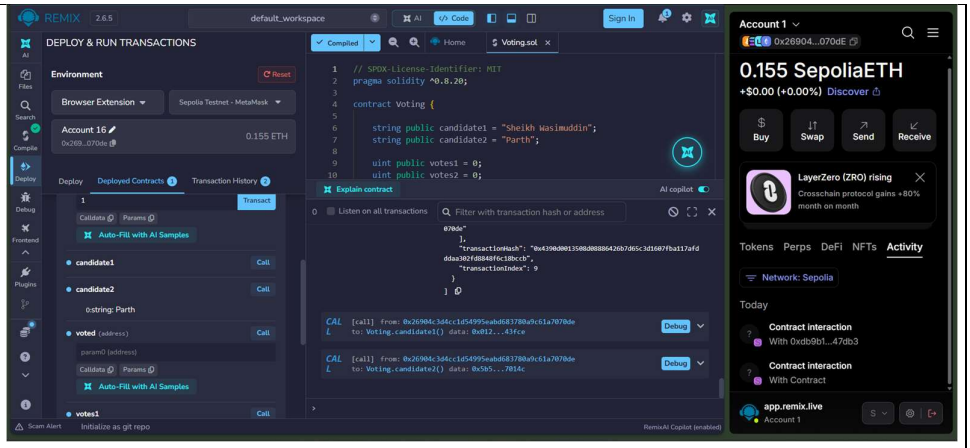


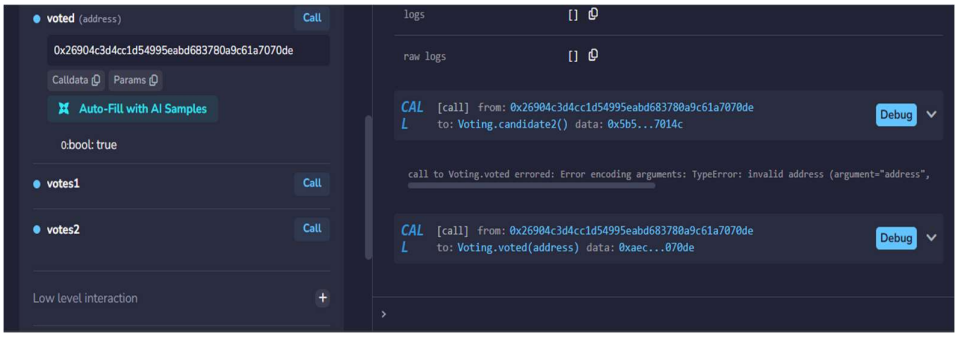
Vision of the Department

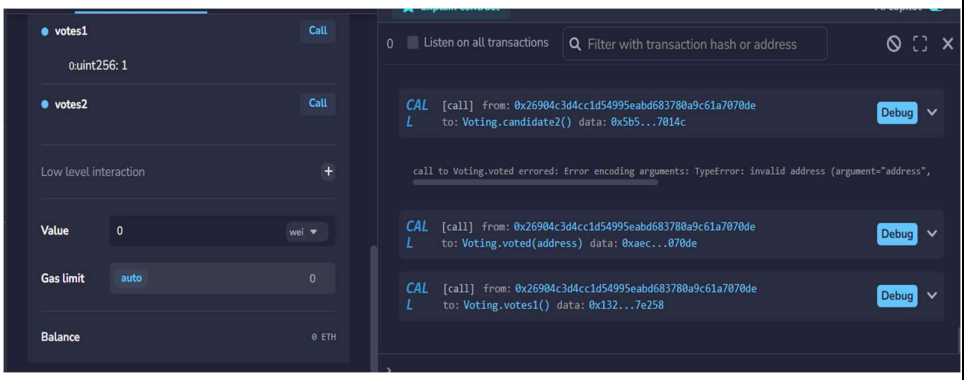
To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

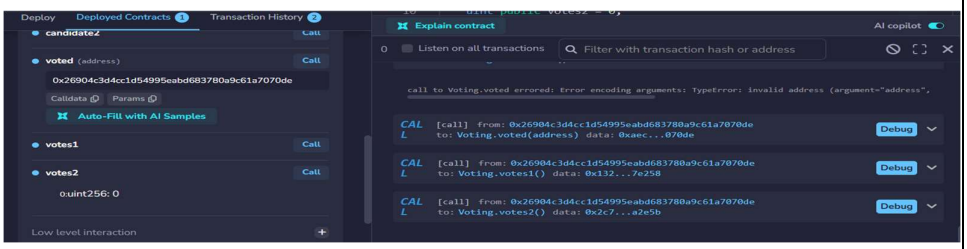
Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.









Code	// SPDX-License-Identifier: MIT
	pragma solidity ^0.8.20;
	
	contract Voting {
	
	string public candidate1 = "Sheikh Wasimuddin";
	string public candidate2 = "Parth";
	
	uint public votes1 = 0;
	uint public votes2 = 0;
	
	mapping(address => bool) public voted;
	
	function vote(uint candidate) public {
	
	require(voted[msg.sender] == false, "Already voted");
	
	if (candidate == 1) {
	votes1++;
	}
	else if (candidate == 2) {
	votes2++;
	}
	else {
	revert("Invalid candidate");
	}
	
	voted[msg.sender] = true;
	}
	}

Conclusion	Thus, the voting smart contract was successfully created using **Solidity** and deployed on the **Sepolia test network** with the help of **Remix IDE and MetaMask**. Sepolia test ETH was obtained through a faucet to pay the required gas fees. The contract allows users to vote for candidates and ensures that one wallet address cannot vote more than once. The votes were successfully recorded and verified on the blockchain, demonstrating the basic working of a secure and transparent blockchain-based voting system.
Date	06/10/2026